

Auteur: Luko De Vriendt
 Studierichting: Informatica
 Oefeningenreeks 5B nr. 2.1

$$\begin{aligned}
 & \int_{-\infty}^{+\infty} \frac{dx}{1+x^2} \\
 &= \lim_{a \rightarrow -\infty} \left(\int_a^0 \frac{dx}{1+x^2} \right) + \lim_{b \rightarrow +\infty} \left(\int_0^b \frac{dx}{1+x^2} \right) \\
 &= \lim_{a \rightarrow -\infty} \left([\text{Bgtan}(x)]_a^0 \right) + \lim_{b \rightarrow +\infty} \left([\text{Bgtan}(x)]_0^b \right) \\
 &= \lim_{a \rightarrow -\infty} (\text{Bgtan}(0) - \text{Bgtan}(a)) + \lim_{b \rightarrow +\infty} (\text{Bgtan}(b) - \text{Bgtan}(0)) \\
 &= - \left(\frac{-\pi}{2} \right) + \frac{\pi}{2} \\
 &= \pi
 \end{aligned}$$

References